# CCSA104 Practical Assignment Report

## Managing Security Policies

*Course: CCSA104 Security Policy Management (2 credit units) · Companion to Lab 4A: Manage the Security Policies · Check Point Certified Security Administrator (CCSA), Cohort 11, School of Professional Development*

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| **Date** | 08 November 2026 | **Score** | / 100 |

## 1. Overview and Lab Environment

This report documents my work on the CCSA104 practical assignment. In Lab 4A a Security Policy is created in SmartConsole, stored on the Security Management Server and enforced by the Security Gateway. Here I applied the same workflow on my own: I examined the default policy, built and organised an Access Control rulebase that follows the Positive Control (whitelist) model, configured Hide NAT, and ran a rule-order experiment while publishing and installing the policy. Part B contains the theory answers.

**Environment.** All tasks were carried out in SmartConsole, logged in as cpadmin, connected to the primary Security Management Server A-SMS (10.1.1.101). The policy package edited throughout is Standard. Every screenshot below is labelled with the task it supports.

**Lab objects.** The assignment says to use the addressing and gateway objects of my own lab topology. The table shows which lab object stands in for each assignment object.

| Assignment object | Object used in my lab | Where it appears |
|---|---|---|
| Internal_LAN (Network) | A-INT-NET / InternalNet | Rules 5 and 6; NAT |
| Admin_PC (Host) | Alpha-Admins | Rule 1 (Admin Access) |
| Gateway | A-GW-01, A-GW-02, A-GW-Cluster | Rules 1 and 2 |
| Web_Server (Host) | Student-Web-Server | Rules 3 and 4 |
| External_DNS (Host) | Azure-External DNS | Rule 5 (DNS Out) |

## 2. Part A: Practical Tasks

### Task 1: Explore the default policy

**Policy package in use.** The package open in SmartConsole is Standard (shown on the tab above the rulebase). Under Manage policies and layers two packages exist: Alpha-Standard (Access Control and Threat Prevention, installed on A-GW-Cluster) and Standard (Access Control and Threat Prevention, installed on all gateways) (Figure 5).

**Default Cleanup rule values.**

| Field | Value |
|---|---|
| **Name** | Cleanup rule |
| **Source** | Any |
| **Destination** | Any |
| **Services & Applications** | Any |
| **Action** | Accept |
| **Track** | Log |
| **Install On** | Policy Targets |

**Showing and hiding columns.** Right-clicking a column header opens the list of available columns: Hits, Name, Source, Destination, VPN, Services & Applications, Action, Time, Track, Install On and Comments. Hits, Time and Comments were not ticked at first (Figure 1). Ticking Hits adds the Hits column to the rulebase (Figure 2), and Track and Install On are visible on the right-hand side (Figure 3). Clearing a tick hides a column and ticking it again restores it.

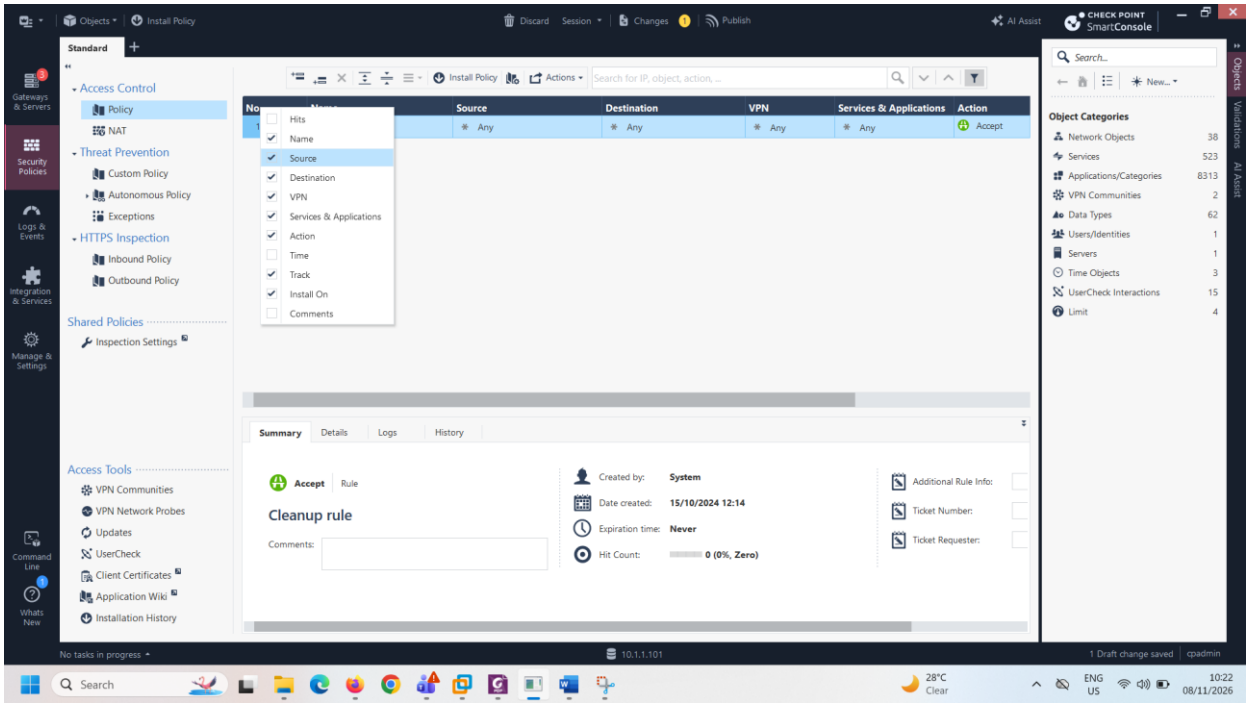



*Figure 1: Task 1 – Right-click column menu on the rulebase header*

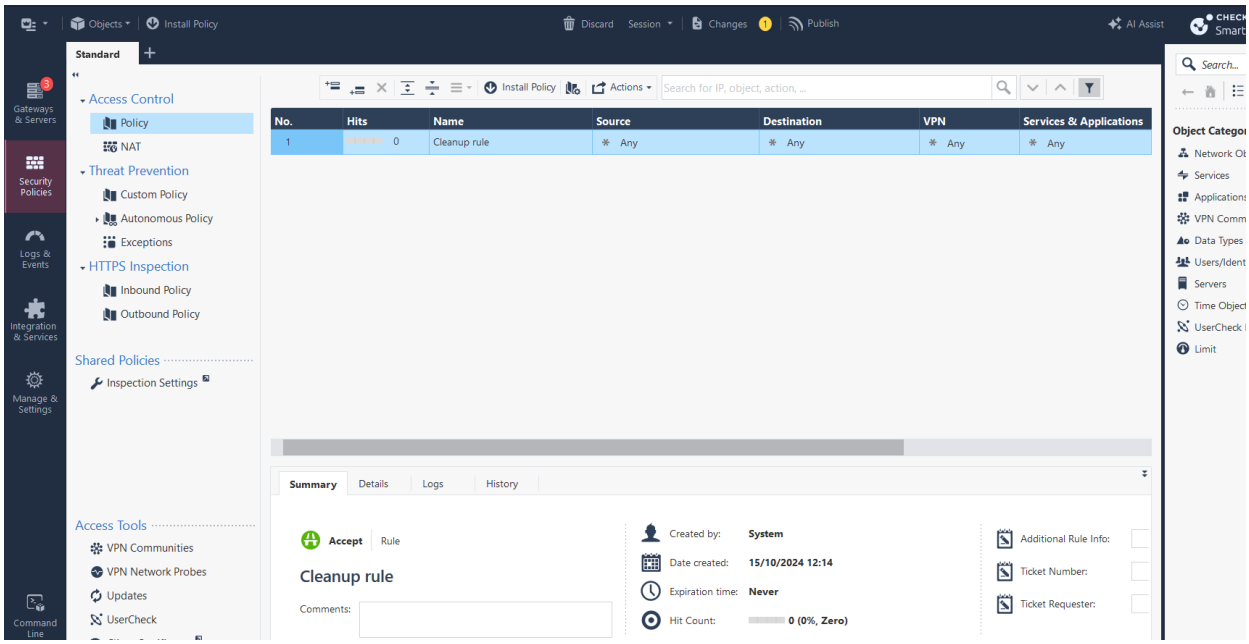



*Figure 2: Task 1 – Default rulebase with the Hits column visible*

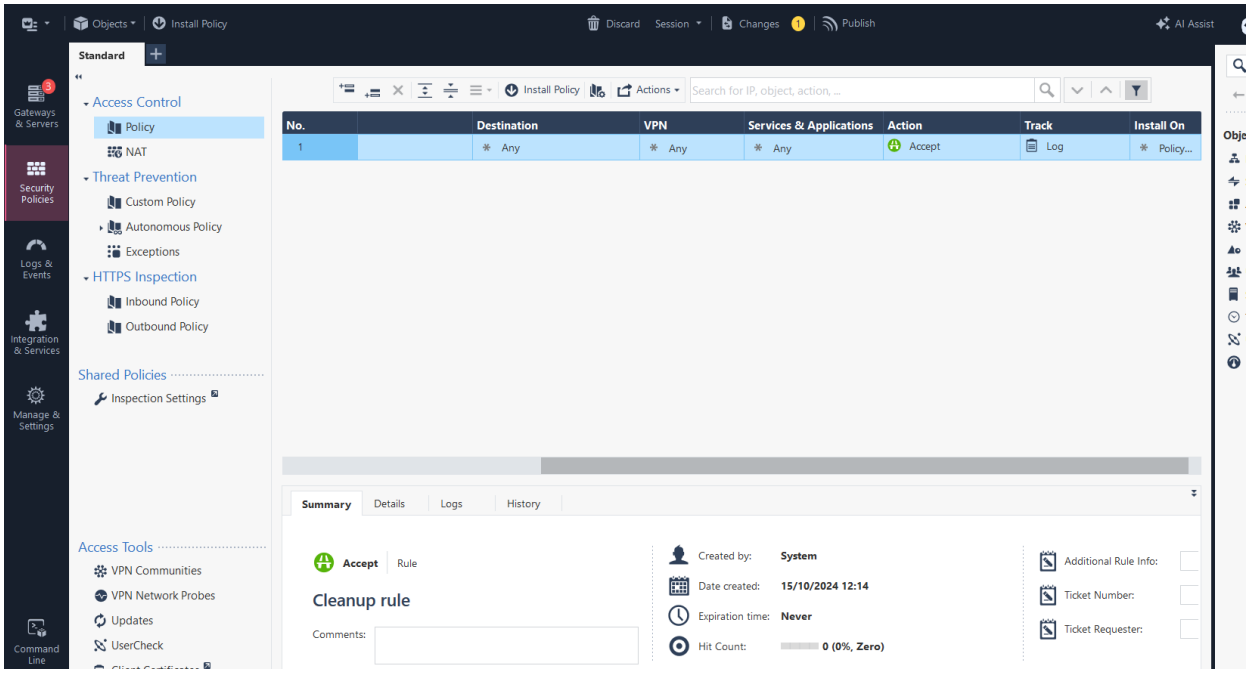



*Figure 3: Task 1 – Default rulebase scrolled to show Track (Log) and Install On (Policy Targets)*

**Global Properties.** Opened from the Applications menu, Global Properties shows a tree of setting categories. Three of them are FireWall (implied rules and their position in the rulebase, plus the Log Implied Rules option), NAT – Network Address Translation, and Log and Alert. Others visible include Authentication, VPN, Identity Awareness, Stateful Inspection, IPS and Hit Count (Figure 4).

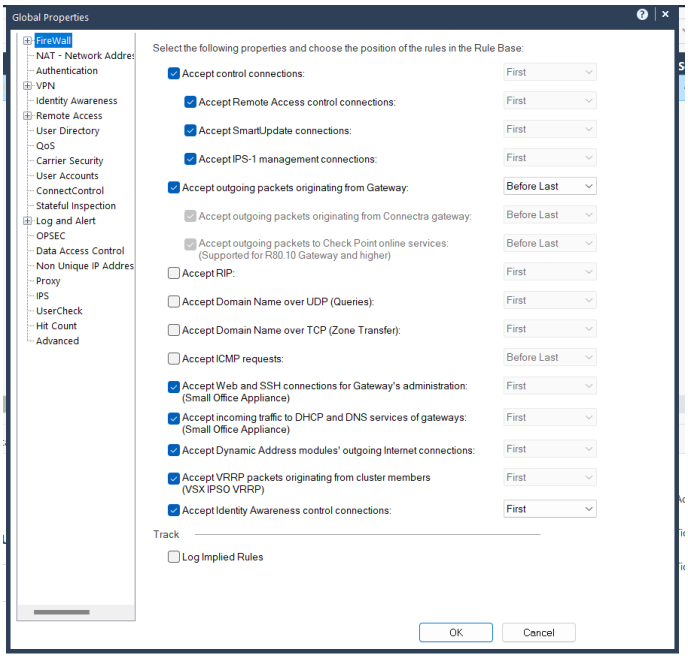



*Figure 4: Task 1 – Global Properties, FireWall page*

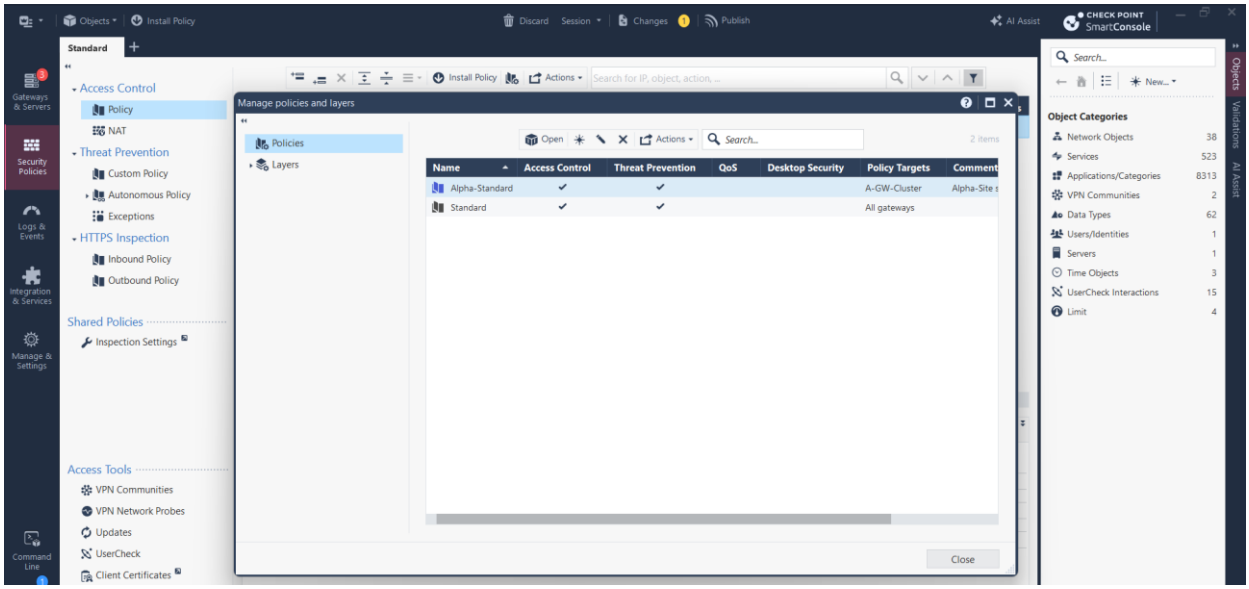



*Figure 5: Task 1 – Manage policies and layers: Alpha-Standard and Standard*

## Task 2: Build the Access Control rulebase

Following the Positive Control Model (allow only what is needed, drop everything else and log it), I inserted the rules above the default Cleanup rule, which stays at the bottom. The management rule comes first, the Stealth rule follows immediately, then the DMZ rules, then the outbound rules. Every rule has Track set to Log so that matches are recorded.

| No. | Name | Source | Destination | Services & Applications | Action | Track |
|---|---|---|---|---|---|---|
| 1 | Admin Access | Alpha-Admins | A-GW-01<br>A-GW-02 | https<br>ssh_version_2 | Accept | Log |
| 2 | Stealth Rule | Any | A-GW-Cluster | Any | Drop | Log |
| 3 | Inbound DMZ | Any | Student-Web-Server | https<br>http<br>smtp | Accept | Log |
| 4 | Internal DMZ | Any | Student-Web-Server | http<br>https | Accept | Log |
| 5 | DNS Out | A-INT-NET | Azure-External DNS | dns | Accept | Log |
| 6 | Internet Webout | InternalNet | Any | http<br>https | Accept | Log |
| 7 | SMTP Out | Any | Any | smtp | Accept | Log |
| 8 | Cleanup rule | Any | Any | Any | Accept | Log |

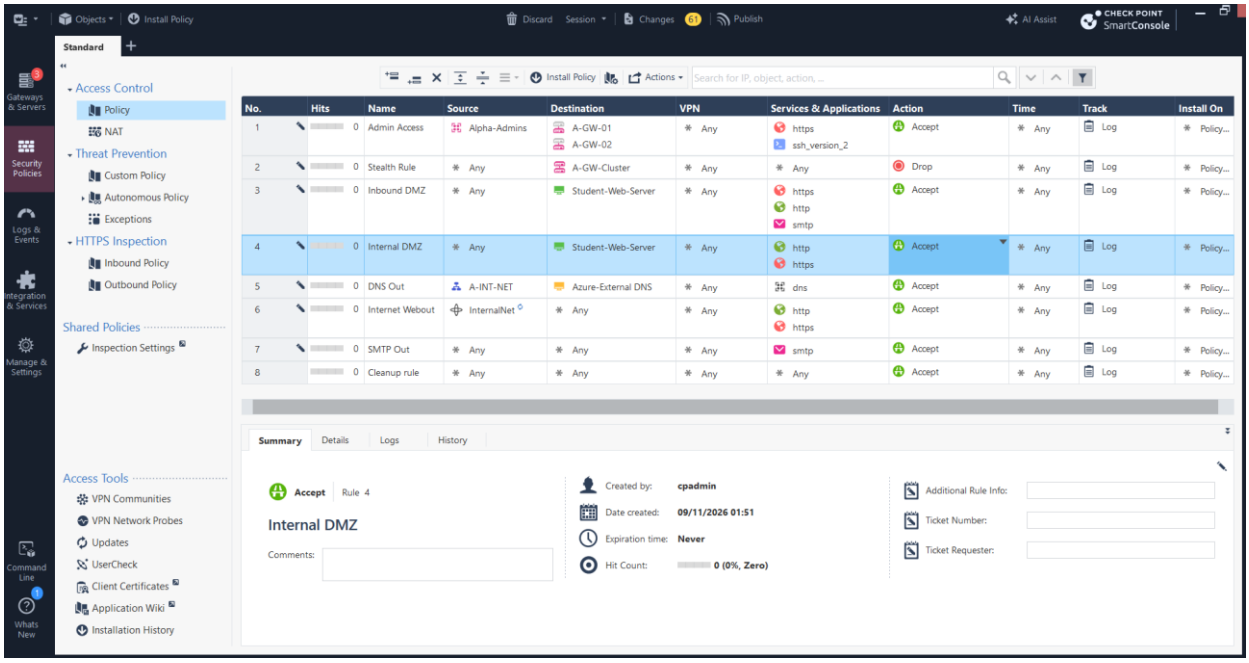



*Figure 6: Task 2 – Rulebase showing number, name, source, destination, services, action and track*

## Task 4: Organise the rulebase with sections

Using New Section Title on the rulebase, I created four sections and placed each rule in the correct one.

| Section title | Rules it contains |
|---|---|
| **Management and Protection** | Rules 1 and 2 (Admin Access, Stealth Rule) |
| **DMZ Services** | Rules 3 and 4 (Inbound DMZ, Internal DMZ) |
| **Outbound Access** | Rules 5, 6 and 7 (DNS Out, Internet Webout, SMTP Out) |
| **Cleanup** | Rule 8 (Cleanup rule) |

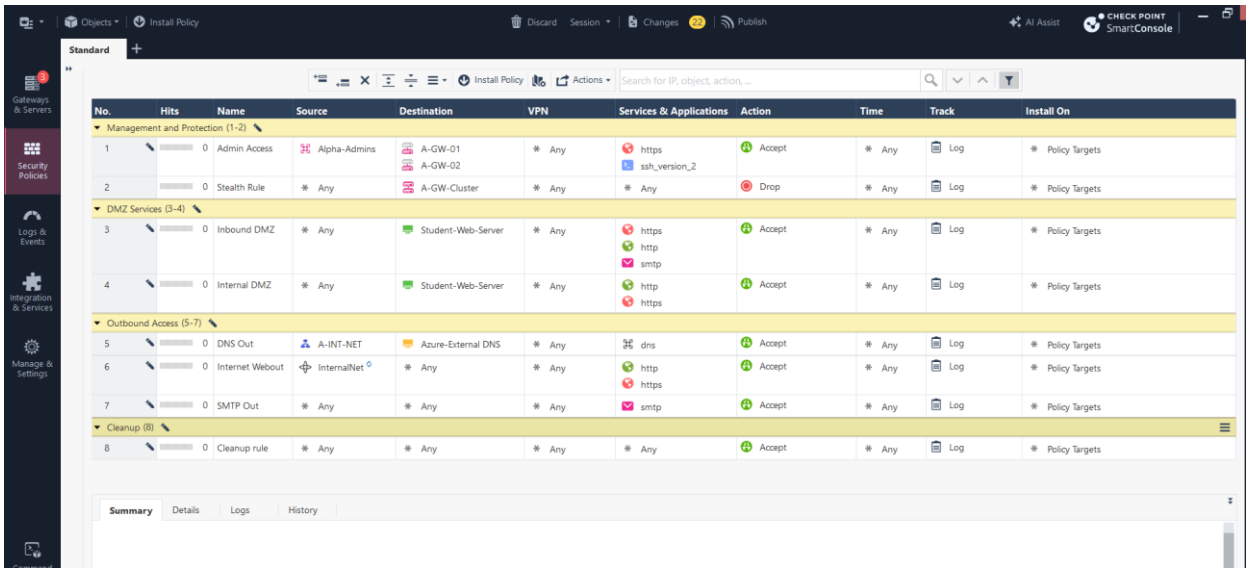



*Figure 7: Task 4 – All four sections expanded*

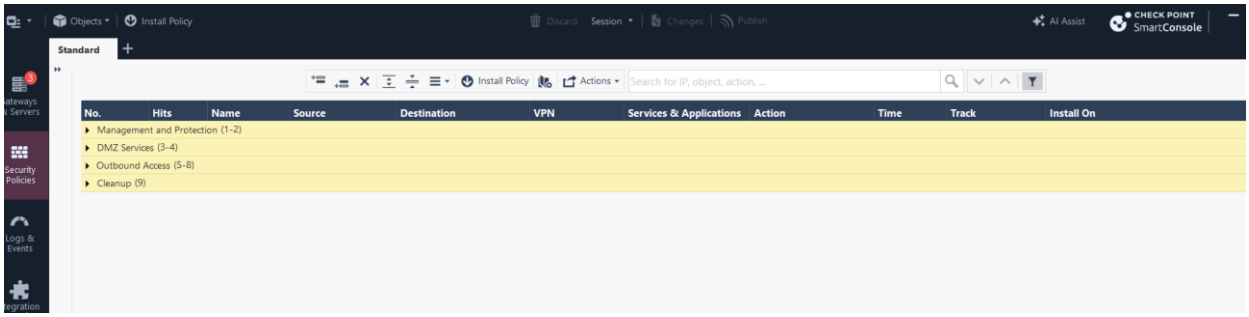



*Figure 8: Task 4 – All sections collapsed (captured while the temporary rule from Task 7 was in the rulebase, so Outbound Access shows 5–8 and Cleanup shows 9)*

**Do sections affect enforcement?** No. Sections are only a visual way of grouping rules: the gateway still enforces the rules in order of their numbers from top to bottom, and the section titles are not sent to the Security Gateway as part of the enforced policy.

## Task 6: Configure Hide NAT

I opened the internal network object (A-INT-NET, "Alpha-Site Internal Network"), went to its NAT page, ticked Add automatic address translation rules, set the Translation method to Hide, chose Hide behind the gateway, left Install on gateway at All, and clicked OK (Figure 9). I then published the change in its own session. The revision list shows it as the "Nat internal netw…" revision with 62 changes (Figure 19). After publishing I installed the database; the Install database dialog listed A-SME (10.1.1.130, Alpha-Site Log Server) and A-SMS (10.1.1.101, primary Security Management), and the Recent Tasks list reports that the installation succeeded for both (Figures 10 and 11).

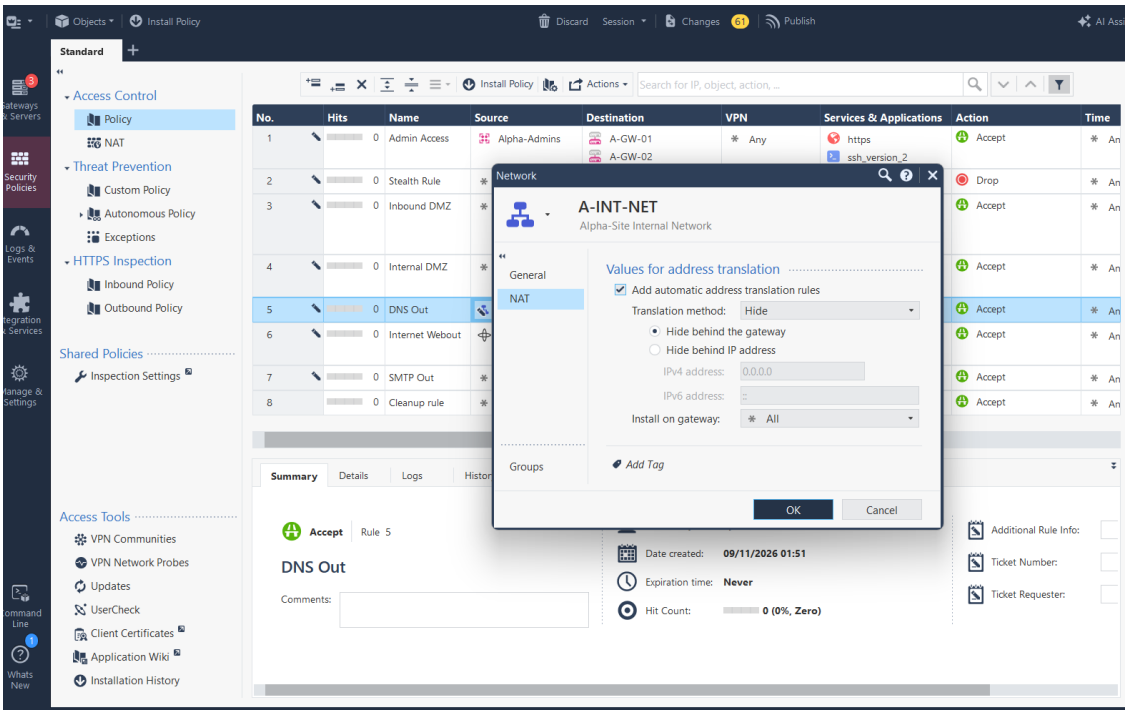



*Figure 9: Task 6 – A-INT-NET NAT page: automatic rules, Hide, behind the gateway*

**INSERT SCREENSHOT:** Task 6 – NAT rulebase (Security Policies > Access Control > NAT) showing the automatic rule(s) for the internal network.

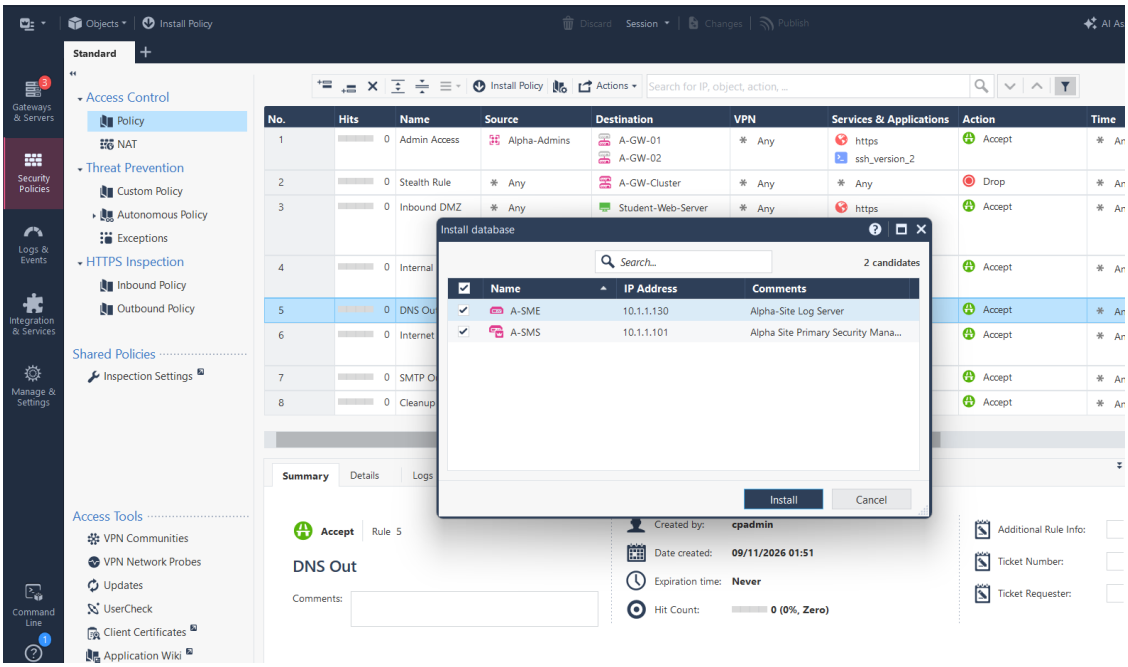



*Figure 10: Task 6 – Install database dialog after publishing the NAT change*

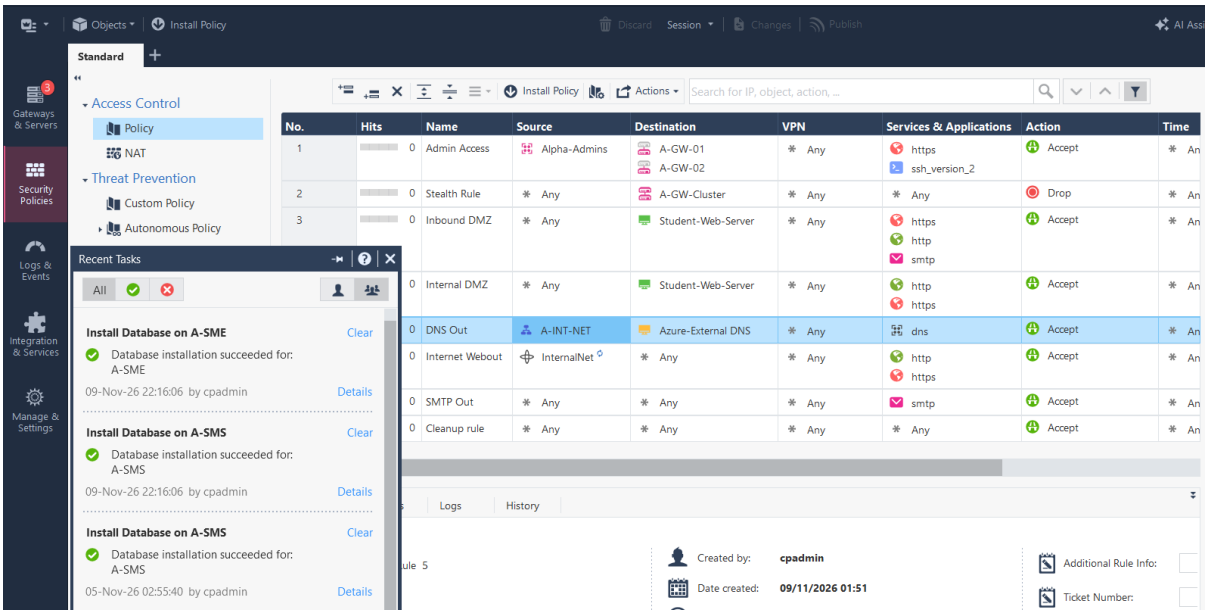



*Figure 11: Task 6 – Recent Tasks: database installation succeeded for A-SME and A-SMS*

**Why this is an automatic NAT rule.** I did not write the rule by hand in the NAT rulebase. I switched on automatic address translation from the network object itself, and SmartConsole generated the matching NAT rule(s) from that object's settings. Automatic rules are tied to the object, so they change or disappear when the object's NAT settings change. A manual rule, by contrast, is created directly in the NAT rulebase, where I choose the original and translated source, destination and service myself.

**Many-to-one translation.** Hide NAT translates many internal hosts to one address. Every host in the internal network, each with its own private address, leaves through the single address of the gateway, so outside systems only ever see that one address and the internal addressing stays hidden.

**What the gateway does with the source port.** Because many hosts share one address, the address alone cannot identify a connection. The gateway therefore replaces the original source port with a unique port for each connection and records the mapping in its connection table. When the reply returns to that port, the gateway translates it back to the correct internal host and original port, so connections from different hosts do not get mixed up.

## Task 7: Rule order experiment and Installation History

### Step 1: TEMP Block Internal above Internet Webout

I added a temporary rule named TEMP Block Internal (Source Any, Destination Any, Services Any, Action Drop, Track Log) in the Outbound Access section, above Internet Webout, so that it became rule 6 and Internet Webout moved to rule 7. I published it in a session named "TEMP Block Internal" (28 draft changes) and installed the database. The installation succeeded for A-SME and A-SMS (Figures 12 to 14).

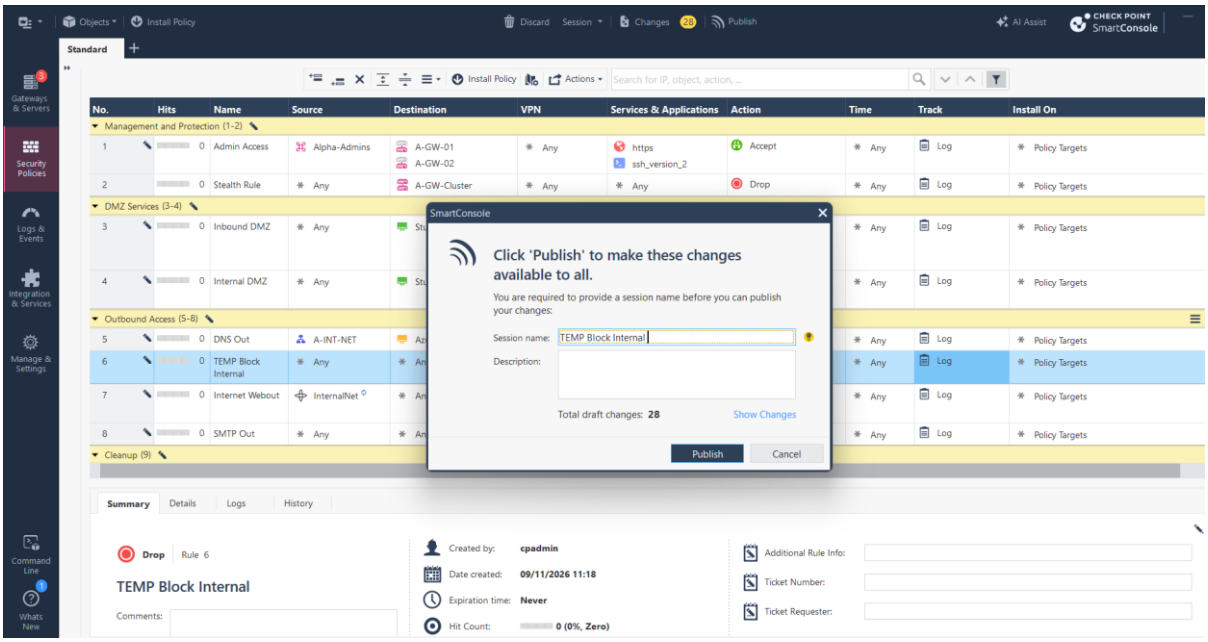



*Figure 12: Task 7 – Publishing TEMP Block Internal (above Internet Webout)*

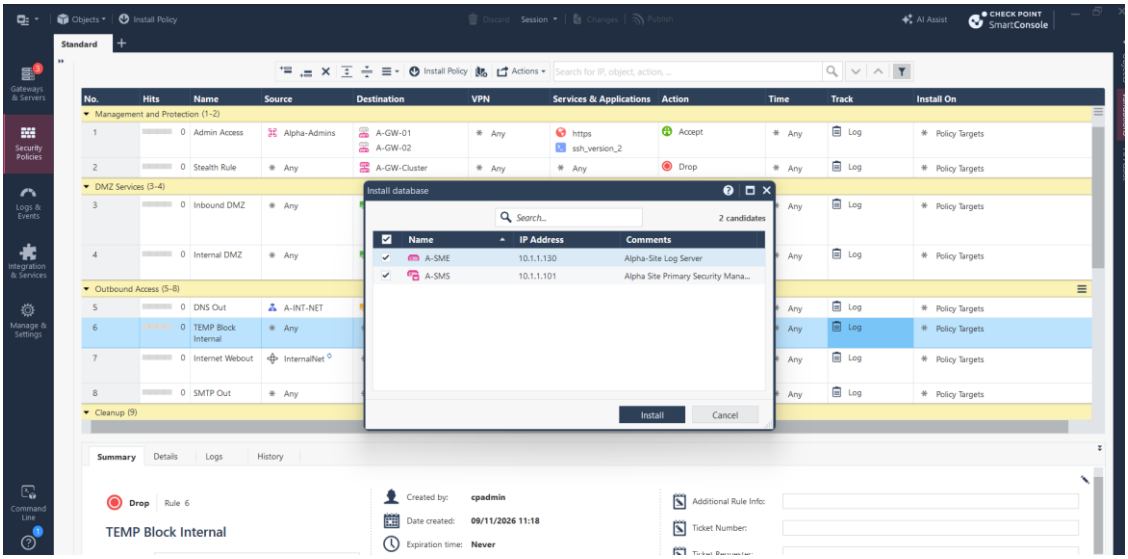



*Figure 13: Task 7 – Install database dialog (A-SME and A-SMS)*

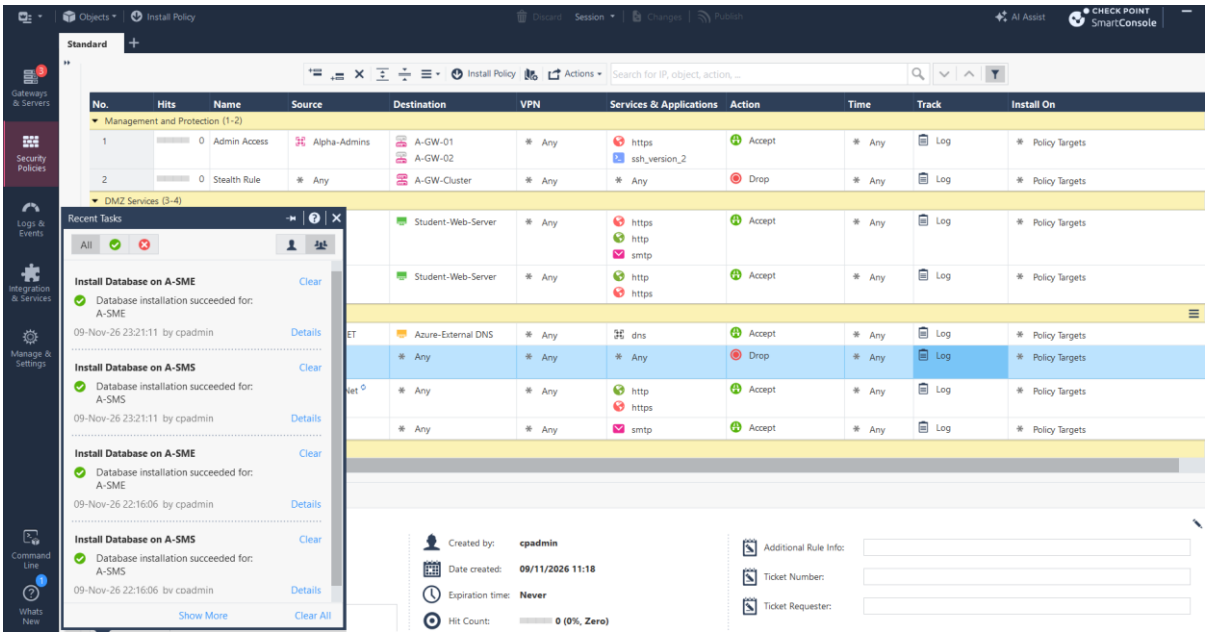



*Figure 14: Task 7 – Install succeeded; TEMP Block Internal (Drop, Log) sits above Internet Webout*

### Step 2: TEMP Block Internal below Internet Webout

I moved TEMP Block Internal below Internet Webout, so Internet Webout became rule 6 and the temporary rule became rule 7. I published this in a session named "TEMP Block Internal below Internal Web Out" (2 draft changes) (Figure 15).

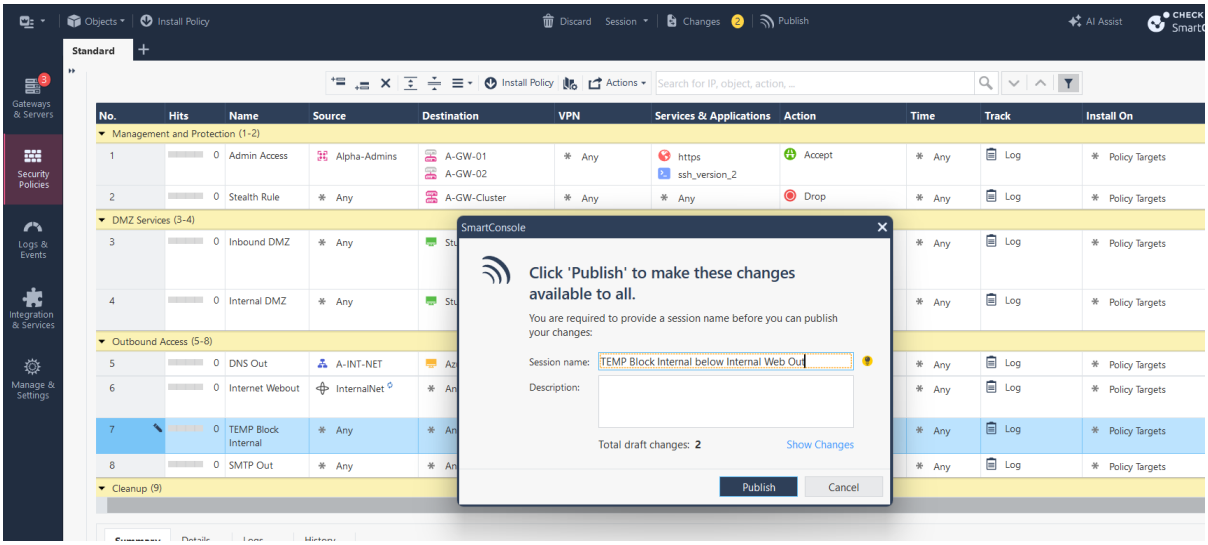



*Figure 15: Task 7 – Publishing after moving TEMP Block Internal below Internet Webout*

**Difference.** The rulebase is read from top to bottom and the first matching rule decides the outcome. With TEMP Block Internal above Internet Webout, outbound traffic matches the broad Drop rule first, so the web Accept rule below it never gets a chance. With TEMP Block Internal below Internet Webout, web traffic matches the specific Accept rule first, and the broad Drop rule only catches whatever is left over.

### Step 3: Delete the temporary rule

I deleted TEMP Block Internal after confirming the prompt, published the change in a session named "Delete the temporary rule, then publish" (1 draft change), and installed again. The Recent Tasks list shows the installation succeeding for A-SME and A-SMS at 23:59:02 (Figures 16 to 18).

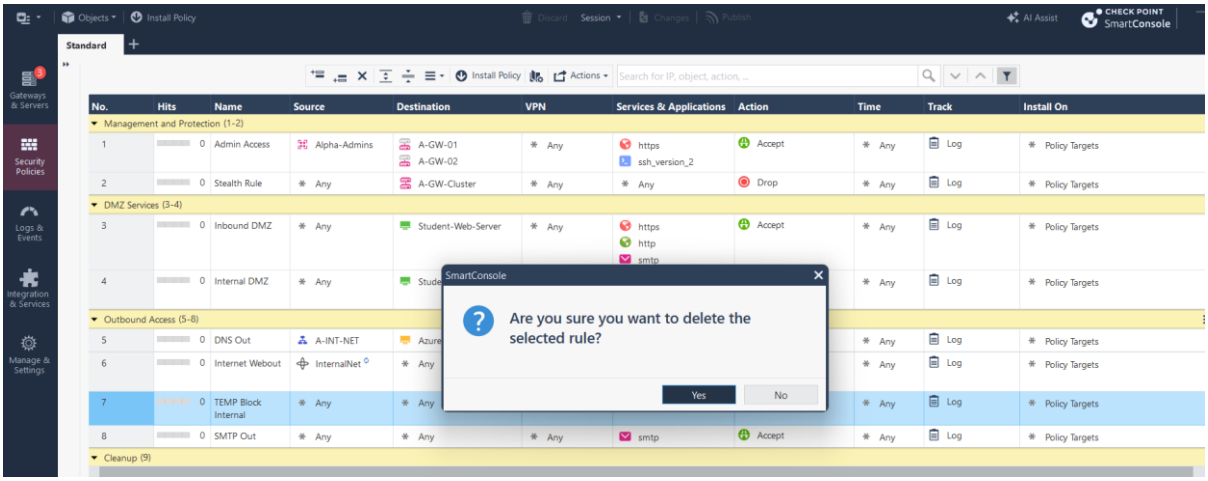



*Figure 16: Task 7 – Confirming deletion of the temporary rule*

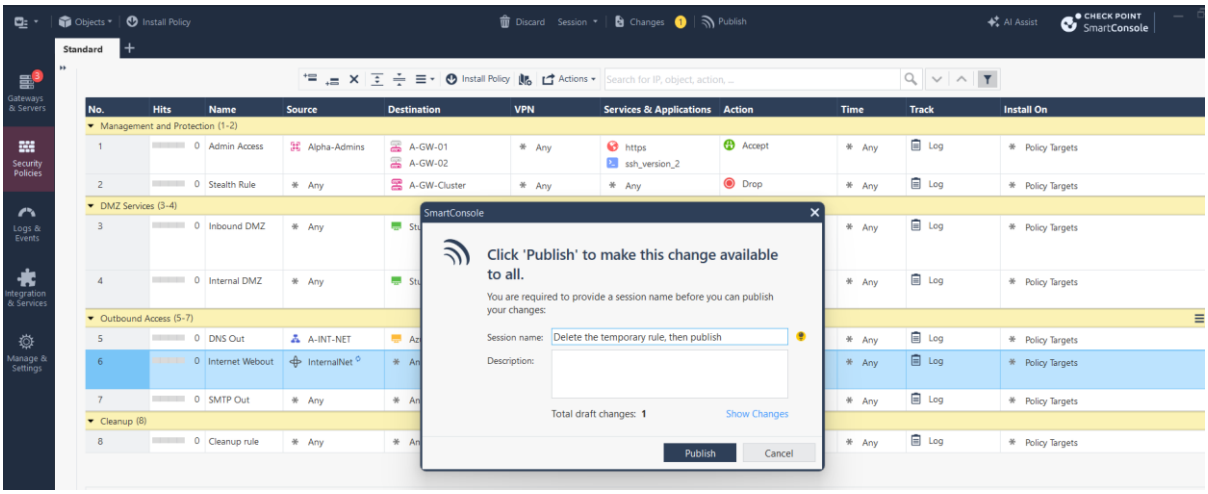



*Figure 17: Task 7 – Publishing the deletion*

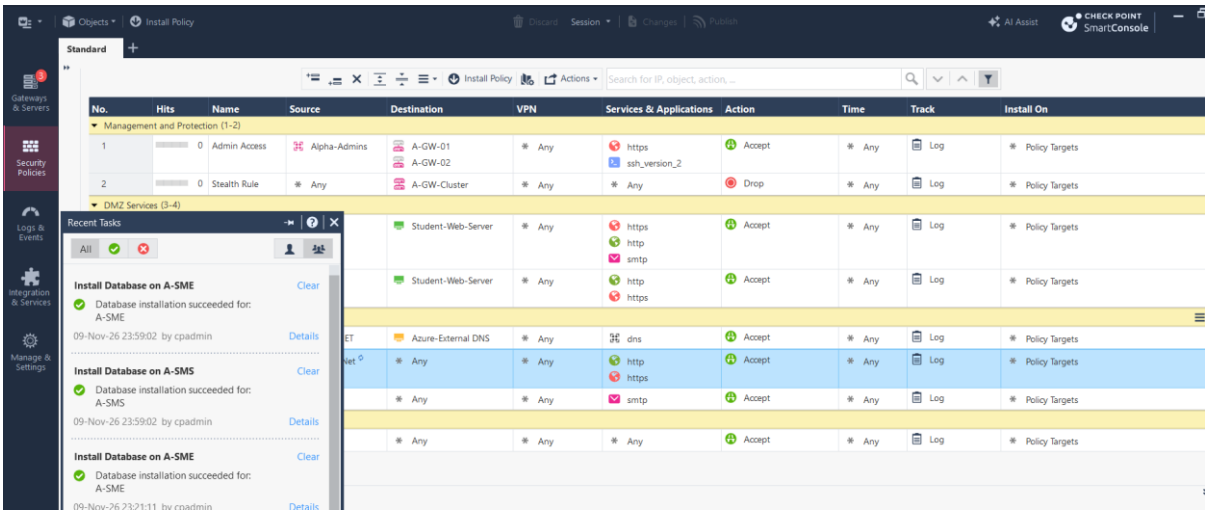



*Figure 18: Task 7 – Install succeeded after the temporary rule was deleted*

### Installation History and audit view

In the revisions view I selected the revision published at 23:31 that contained the temporary rule. Its Audit Logs tab shows two entries by cpadmin: a Modify Rule operation on the Access Control Rule "TEMP Block Internal" and the Publish that committed it (Figure 19). The Network layer history lists the sequence of revisions: the NAT change, the three TEMP Block Internal publishes, and finally "Delete the temporary rule, then publish" (Figure 20). I only viewed these entries; I did not revert anything.

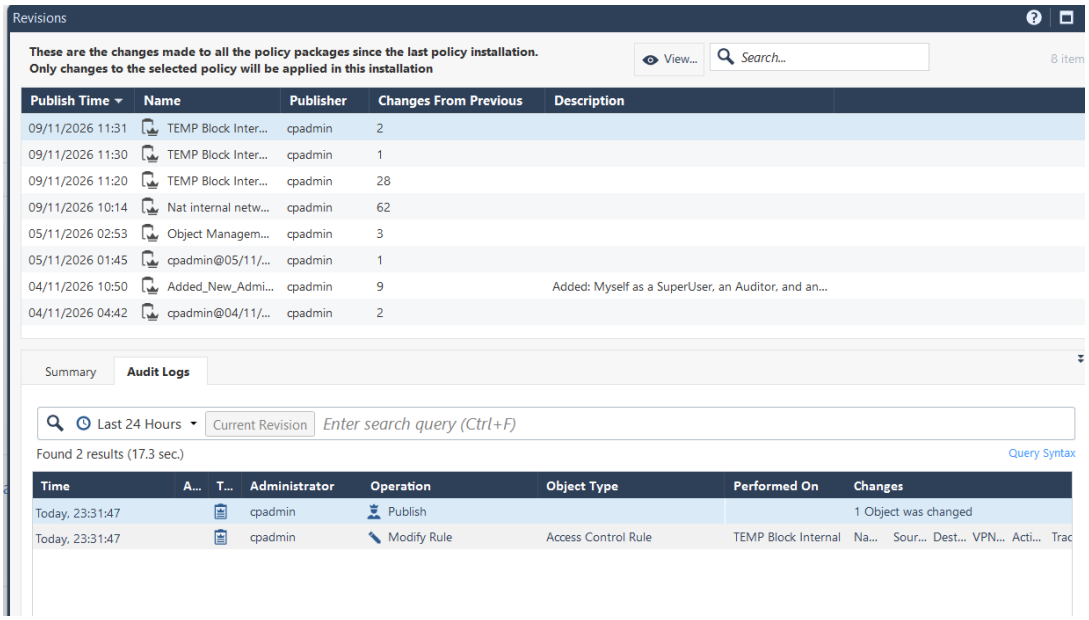



*Figure 19: Task 7 – Revisions view with Audit Logs for the revision containing TEMP Block Internal*

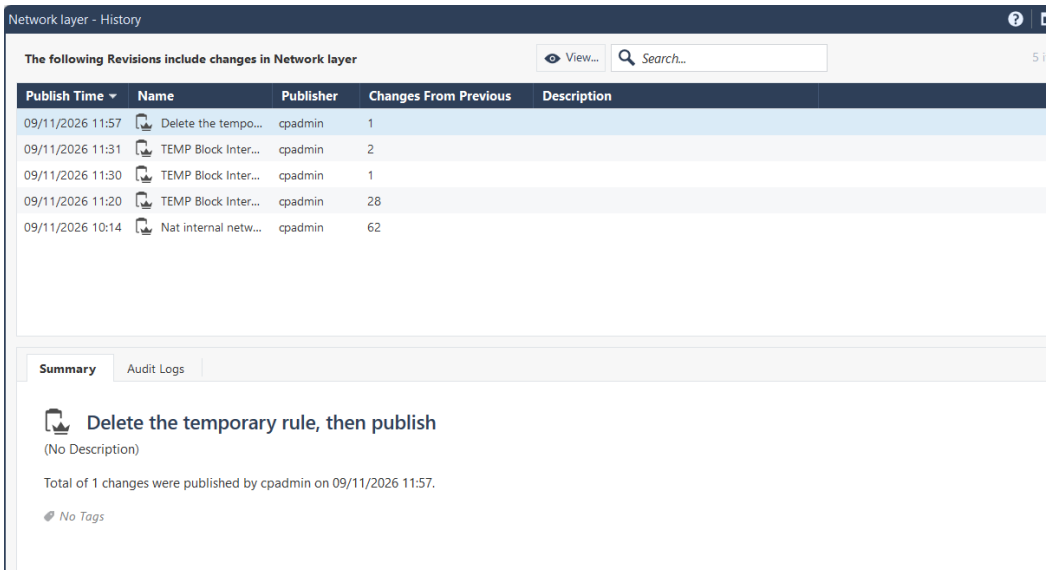



*Figure 20: Task 7 – Network layer history ending with the deletion of the temporary rule*

**Why specific rules belong above general rules.** Check Point reads the rulebase from top to bottom and applies the first rule that matches a connection, ignoring every rule below it. A specific rule such as Internet Webout must therefore sit above a general rule such as TEMP Block Internal, otherwise the general rule captures the traffic first and the specific permission never takes effect, as the first test showed. Keeping specific rules at the top and general rules near the bottom lets precise permissions work as intended while the broad rules catch only what remains.

# 3. Part B: Theory Questions

## Section B1: Multiple choice

| Question | Answer | Reason |
|---|---|---|
| **Q1** | B | The policy is created in SmartConsole, stored on the Security Management Server and enforced by the Security Gateways. |
| **Q2** | A | The Access Control policy follows the Positive Control (whitelist) model: only what is explicitly allowed passes and everything else is dropped. |
| **Q3** | C | The Stealth rule should sit as early as possible, immediately after the Management rules, so nothing else can reach the gateway directly. |
| **Q4** | D | Sections are visual divisions only and do not change the order in which rules are enforced. |
| **Q5** | B | Hide NAT translates many internal hosts to one address (many-to-one), using different source ports to tell the connections apart. |

## Section B2: Short answers

### Q6. Explicit rules, Implied rules and the Implicit Cleanup Action

Explicit rules are the rules an administrator writes in the Access Control rulebase; they are visible there with a number and can be edited, moved and logged. Implied rules are predefined by Check Point so that the gateway keeps working, for example accepting control connections or outgoing packets that originate from the gateway, and they are not shown in the rulebase by default. Implied rules are configured in Global Properties on the FireWall page (Figure 4), and each one can be placed First, Before Last (just before the Cleanup rule) or Last. The Implicit Cleanup Action is the built-in final action applied to traffic that matches no explicit or implied rule; for the Access Control policy this is Drop.

### Q7. Why rule order matters

The gateway checks a connection against the rules from top to bottom and acts on the first match, so the order decides the outcome. If a broad rule sits above a narrow one, it captures the traffic and the narrow rule is never reached. Example from my rulebase: Admin Access (rule 1) must stay above the Stealth Rule (rule 2); if the Stealth Rule came first, it would drop the administrators' own connections to the gateway. In Task 7 the same principle showed up when TEMP Block Internal, placed above Internet Webout, blocked the web traffic that Internet Webout was meant to allow.

### Q8. Saving versus publishing, and installing

Saving keeps my changes in my own private draft session, so only I can see them (the status bar shows "Draft changes saved"). Publishing makes the changes available to all administrators and commits them to the management database. When a policy is installed, only published changes are included; unpublished changes are not part of the installation. Many organisations publish only during a change window because published changes become visible to everyone and are carried by the next installation, so publishing in a controlled window prevents half-finished or unreviewed changes from reaching production by accident.

### Q9. Global Properties compared with Policy Packages

Global Properties holds settings that apply across the whole management environment, independent of any single policy. For example, the FireWall page controls the implied rules and where they are placed. I opened it from the Applications menu by choosing Global Properties (Figure 4). A Policy Package is a named set of policies (Access Control, Threat Prevention and so on) that is installed on chosen targets. For example, the Alpha-Standard package targets A-GW-Cluster while Standard targets all gateways. I opened the list from the Applications menu by choosing Manage policies and layers (Figure 5).

### Q10. Security Zones

The four predefined Security Zones are WirelessZone, ExternalZone, DMZZone and InternalZone. Two rules for assigning gateway interfaces: an interface can belong to only one zone at a time, and one zone can contain several interfaces. A benefit of zone-based rules is that a rule written for a zone keeps working when networks behind the interfaces change, so there are fewer rules to write and maintain than when every network is listed individually.

## Section B3: Scenario questions

### Q11. Accept-all rule above the Stealth rule

**Two problems.** First, an Any/Any/Any Accept rule above the Stealth rule matches every connection, including connections to the gateway itself, so the Stealth rule never matches and the gateway is exposed to scanning and attacks from anywhere. Second, every rule below the new rule is shadowed and never matched, so the DMZ, outbound and Cleanup rules do nothing; the policy no longer follows the Positive Control Model and traffic that should be dropped and logged is accepted.

**Correction.** Remove the rule, or at least move it below the Stealth rule and narrow it to the specific sources, destinations and services that are really needed, with Track set to Log. Then publish and install the policy and check the hit counts to confirm the other rules match again.

**Rule allowed above the Stealth rule.** Only the Management rule, such as Admin Access, may sit above it. Without it, the Stealth rule would drop the administrators' own connections to the gateway and lock them out. This is safe because the rule is tightly scoped to trusted administrator sources and management services only.

### Q12. Staff cannot reach the Internet after Hide NAT

**(a) Two benefits of NAT.** NAT hides the internal private addressing from the outside world, which reduces what attackers can learn about the internal network. It also allows many hosts to share one public address, saving scarce public IPv4 addresses and avoiding the need to renumber the internal network.

**(b) Three checks to find the fault.**

- **Check the logs.** In Logs & Events I would filter on the internal source address to see whether the traffic is accepted, dropped or not seen at all, and which rule number matched. This shows immediately whether a rule such as Internet Webout or the Cleanup rule is affecting the traffic.
- **Check the NAT rule and the installation.** I would confirm that the automatic Hide NAT rule exists in the NAT rulebase, that the change was published, that the policy was installed on the correct gateway (Install on gateway), and that the NAT settings are set to Hide behind the gateway. An unpublished or uninstalled change would leave the old behaviour in place.
- **Check the gateway interfaces, routing and DNS.** I would verify that the external interface has a valid address and a default route towards the Internet, and that DNS works for the clients. Hide NAT only translates the address; if the route or DNS is broken, staff still cannot reach the Internet even though the policy is correct.